

Report
Of
Data Analytics with Python & Power BI

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
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Batch: 2026–27

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PREFACE

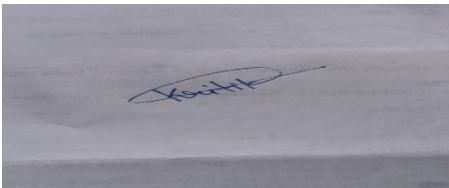
This report presents the learning outcomes, practical experience, and project work completed during the AICTE–EduSkills Virtual Internship on Data Analytics with Python & Power BI. It summarizes the concepts learned, tools used, methodologies followed, and the implementation of the capstone project. The report has been prepared as part of the academic requirements of the B.Tech Computer Science & Engineering program at IILM University and reflects the knowledge and skills gained during the internship.

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the AICTE–EduSkills Virtual Internship Program on Data Analytics with Python & Power BI. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to the mentors and facilitators at EduSkills Academy for sharing their truthful and illuminating views on the concepts covered during the internship.

I would also like to thank my faculty coordinator at IILM University, Greater Noida, for continuous support and guidance in the successful completion of this internship and this report.

Finally, I express my gratitude to my parents for their blessings and constant encouragement throughout this journey.

A photograph of a handwritten signature in blue ink on a white piece of paper. The signature appears to be 'Kritik Kumar'.

Student Name: Kritik Kumar

Roll No.: CS23411627

INTERNSHIP COMPLETION CERTIFICATE

This internship (June 2026 – October 2026) is currently in progress. As per the offer letter issued by EduSkills (Ref No. C18Y2026M081555618), the completion certificate is awarded only after successful completion of all weekly assessments, project submissions, and the Final Assessment Test in the last week of the program.

The Internship Completion Certificate will be inserted on this page once it is issued by EduSkills at the conclusion of the internship.

Chapter 4: Project Description

4.1 Introduction

In today's data-driven world, organizations across industries rely heavily on structured data analysis to make informed business decisions. The ability to collect, clean, analyze, and visualize data has become one of the most sought-after skill sets in the technology industry. Recognizing this need, the All India Council for Technical Education (AICTE), in collaboration with EduSkills Academy, offers structured Virtual Internship Programs to engineering students under the National Internship Portal, an initiative of the Ministry of Education, Government of India.

This report documents my learning, work, and outcomes during the 8-week AICTE–EduSkills Virtual Internship Program undertaken in the domain of Data Analytics with Python & Power BI, as part of the B.Tech (Computer Science & Engineering) curriculum at IILM University, Greater Noida. The internship spans the period of August 2026 to October 2026 and follows a structured, week-wise learning plan combining conceptual learning with hands-on, industry-relevant practice.

The purpose of this report is to present a comprehensive account of the internship experience — covering the organization under which the internship was conducted, the problem addressed through the capstone project, the tools and technologies used, the system architecture followed, the methodology adopted on a week-by-week basis, and the outcomes and learnings achieved by the end of the program.

4.2 Organization Profile

The internship was conducted under the aegis of EduSkills Academy, a skilling organization working under the patronage of AICTE and the Ministry of Education's National Internship Portal. EduSkills partners with academic institutions and industry to design and deliver structured Virtual Internship Programs that are aligned with current industry requirements, aiming to enhance the employability and technical readiness of engineering students across India.

EduSkills Academy is certified under ISO 9001:2015 (Quality Management) and ISO 21001:2018 (Educational Organizations Management) and has been recognized as a Great Place to Work. The Data Analytics with Python & Power BI internship track is delivered as an 8-week virtual program consisting of week-wise structured modules,

weekly assessments, project submissions, and a capstone project culminating in a Final Assessment Test.

4.3 Problem Statement

Modern e-commerce businesses generate large volumes of raw transactional data on a continuous basis — covering customer orders, product categories, sales channels, and regional performance. However, this raw data is often unstructured, inconsistent, and scattered across multiple sources, making it difficult for business stakeholders to derive timely and actionable insights.

The capstone project undertaken during this internship addresses this exact problem: given a raw, unclean e-commerce dataset, the objective is to clean and transform the data using Python, extract and analyze it using SQL, and build an interactive Power BI dashboard that identifies key performance indicators (KPIs) and delivers clear, decision-ready business insights to stakeholders.

4.4 Project Objectives

- To gain hands-on proficiency in Python for data cleaning, transformation, and exploratory data analysis.
- To learn and apply SQL for structured data extraction, querying, and aggregation.
- To develop practical skills in Microsoft Power BI for data modeling and interactive dashboard design.
- To integrate Python, SQL, and Power BI into a single, cohesive analytics workflow.
- To apply the above skills on a real-world style e-commerce dataset through a structured capstone project.

4.5 Scope of the Project

The scope of this internship project is limited to the structured curriculum defined by the AICTE–EduSkills program, covering foundational and intermediate concepts of Python, SQL, and Power BI over an 8-week period. The project focuses on data cleaning, exploratory analysis, dashboard design, and insight generation using a sample e-commerce dataset provided as part of the capstone module.

The scope does not extend to production-grade deployment, real-time streaming data pipelines, or integration with live enterprise systems, as the internship is oriented towards skill-building and academic learning rather than enterprise software delivery.

4.6 Technologies and Tools Used

A combination of programming languages, libraries, and business intelligence tools were used throughout the internship, summarized below:

- Python: Used as the primary programming language for data cleaning, transformation, and scripting.
- NumPy: Used for efficient numerical computation and array-based operations on datasets.
- Pandas: Used extensively for data cleaning — handling missing values, removing outliers, and reshaping datasets into analysis-ready formats.
- Matplotlib & Seaborn: Used for exploratory data analysis (EDA) through statistical plots and visualizations.
- SQL: Used for structured data querying, including SELECT, WHERE, GROUP BY, joins, subqueries, Common Table Expressions (CTEs), and window functions.
- Microsoft Power BI: Used for data modeling, DAX-based measures, time intelligence functions, and building advanced interactive visuals such as drill-through and bookmarks.
- Power BI Service: Used for publishing, sharing, and collaborating on completed dashboards within a managed workspace.

4.7 System Architecture

The overall analytics workflow followed during the capstone project can be represented as a linear pipeline, beginning with raw data ingestion and ending with the delivery of interactive dashboards. Figure 4.1 below illustrates this architecture.



Figure 4.1: Data Analytics Workflow Architecture

As shown above, raw e-commerce data first passes through a Python-based cleaning and preprocessing stage, where missing values, duplicates, and inconsistent formats are handled using Pandas and NumPy. The cleaned data is then stored and queried using SQL, enabling structured extraction, aggregation, and joining of relevant tables. The queried data is subsequently loaded into Power BI, where it is modeled using relationships and DAX measures, and finally visualized through interactive dashboards that deliver actionable business insights to stakeholders.

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on practice, and assessments, as summarized in Table 4.1 below.

Week	Module	Description
Week 1	Data Analytics Foundations & Python Basics	Environment setup (Python/Anaconda, SQL, Power BI); core Python concepts; NumPy and Pandas fundamentals.
Week 2	Data Cleaning & SQL Fundamentals	Data preprocessing with Pandas (missing values, outliers, transformations); SQL SELECT, WHERE, GROUP BY, aggregations, joins.
Week 3	Advanced SQL & Power BI Basics	Subqueries, CTEs, window functions, DML operations; introduction to Power BI interface, data connection, basic visuals.
Week 4	Python-SQL Integration & Data Modeling	Python database connections, automation workflows; Power BI data modeling, DAX basics, time intelligence.
Week 5	Advanced Visualization	Advanced Power BI visuals (drill-through, bookmarks, filters); Python visualization with Matplotlib and Seaborn.
Week 6	Integration, Publishing & Collaboration	Python scripting in Power BI; publishing dashboards to Power BI Service; sharing and workspace management.
Week 7	Real-World Projects & Best Practices	Sales performance analysis; customer segmentation; structured workflow best practices.
Week 8	Capstone Project	E-commerce dataset cleaning; SQL-

Week	Module	Description
		based analysis; KPI identification; dashboard build and final presentation.

Table 4.1: Week-wise Internship Structure

Each week built upon the previous one — beginning with Python and data cleaning foundations, progressing through SQL and Power BI fundamentals, and culminating in a capstone project that required applying all previously learned concepts in an integrated, end-to-end analytics workflow. Weekly assessments were used to validate conceptual understanding, while the final two weeks were dedicated to real-world style project work and the capstone deliverable.

4.9 Expected Outcomes

- Hands-on proficiency in the Python–SQL–Power BI analytics stack widely used in the industry.
- Ability to convert raw, inconsistent data into clean, structured, analysis-ready datasets.
- Competence in writing SQL queries for real-world data extraction and aggregation tasks.
- Skill in designing and publishing interactive, decision-ready Power BI dashboards.
- Strengthened structured problem-solving ability and adherence to workflow best practices.
- A completed capstone project demonstrating the end-to-end analytics workflow on an e-commerce dataset.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter issued by EduSkills, confirming enrollment in the program, is attached below as proof of internship. The Internship Completion Certificate and any further communication proof (such as email confirmations) will be added to this section once received at the conclusion of the internship in October 2026.

Ref No: C18Y2026M081555618

Internship Offer Letter

Student Name : Kritik kumar
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Data Analytics with Python & Power BI
Internship Duration : June 2026 - August 2026
AICTE Student ID : TU6849820263b741749647874

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject areas.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

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IP: 3.111.57.217 | Device: Desktop

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Figure 4.2: Internship Offer Letter (Ref No. C18Y2026M081555618)

Chapter 5: Bibliography / References

1. AICTE – EduSkills Virtual Internship Program. EduSkills Foundation. Retrieved from <https://eduskillsfoundation.org>
2. National Internship Portal. All India Council for Technical Education (AICTE), Ministry of Education, Government of India. Retrieved from <https://internship.aicte-india.org>
3. McKinney, W. (2017). Python for Data Analysis (2nd ed.). O'Reilly Media.
4. Pandas Development Team. Pandas Documentation. Retrieved from <https://pandas.pydata.org/docs/>
5. NumPy Developers. NumPy Documentation. Retrieved from <https://numpy.org/doc/>
6. Microsoft Corporation. Power BI Documentation. Retrieved from <https://learn.microsoft.com/power-bi/>
7. Python Software Foundation. Python 3 Documentation. Retrieved from <https://docs.python.org/3/>
8. Waskom, M. (2021). Seaborn: Statistical Data Visualization. Journal of Open Source Software.